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Department of Mathematics
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Math 208 Major Exam I Term 252

Original exam date: 16 February 2026. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. The solution of the system

$$\begin{aligned} x_1 + 3x_2 + 3x_3 + 3x_4 &= 0 \\ 2x_1 + 7x_2 + 5x_3 - x_4 &= 0 \\ 2x_1 + 7x_2 + 4x_3 - 4x_4 &= 0 \end{aligned}$$

is the set of all scalar multiples of a vector $\mathbf{u}$, where $\mathbf{u} = (a, b, c, d)$. Then $\dfrac{a}{c} =$

(a) 4

(b) -4

(c) 0

(d) -2

(e) 2

2. The sum of all real values of k for which the vectors

$$\mathbf{v}_1 = (2, 3, 0), \quad \mathbf{v}_2 = (4, k, 4), \quad \mathbf{v}_3 = (-1, 1, k)$$

do not form a basis for $\mathbb{R}^3$ is

(a) 6

(b) 12

(c) 4

(d) -12

(e) 5

3. A certain radioactive substance decomposes at a rate directly proportional to the quantity of the substance present. If 50% of a certain quantity has decomposed in 20 years, how many years will it take for 80% of the original quantity to decompose?

(a) $\dfrac{20 \ln 5}{\ln 2}$

(b) $\dfrac{15 \ln 5}{\ln 2}$

(c) $\dfrac{24 \ln 5}{\ln 2}$

(d) $\dfrac{30 \ln 5}{\ln 2}$

(e) $\dfrac{40 \ln 5}{\ln 2}$

4. The general solution of the exact differential equation

$$\left(2x + y^2 e^{xy}\right) dx + \left(e^{xy} + xye^{xy}\right) dy = 0$$

is

(a) $ye^{xy} + x^3 = c$

(b) $ye^{xy} + x^2 = c$

(c) $xe^{xy} + x^2 = c$

(d) $xe^{xy} + y^2 = c$

(e) $ye^{xy} + y^2 = c$

5. All values of m such that $y = e^{mx}$ is a solution of the differential equation

$$y'' - 6y' - 7y = 0$$

are

(a) $m = 7$ and $m = -1$

(b) $m = 7$ and $m = 2$

(c) $m = 7$ and $m = 1$

(d) $m = -7$ and $m = 2$

(e) $m = 7$ and $m = -2$

6. The order of the differential equation

$$y''' + 2x(y'')^4 + x^2(y')^6 = 0$$

is

(a) 3

(b) 10

(c) 1

(d) 4

(e) 6

7. The general solution of the differential equation

$$\frac{dy}{dx} = x\sqrt{\frac{1-y^2}{1-x^2}}$$

for $|x| < 1$ and $|y| < 1$ is given by

(a) $y = 2\sin\left(c - \sqrt{1-x^2}\right)$

(b) $y = \frac{1}{2}\sin\left(c - \sqrt{1-x^2}\right)$

(c) $y = \frac{1}{3}\sin\left(c - \sqrt{1-x^2}\right)$

(d) $y = \sin\left(c - \sqrt{1-x^2}\right)$

(e) $y = 3\sin\left(c - \sqrt{1-x^2}\right)$

8. By using a suitable substitution, we can transform the differential equation

$$y' + \frac{2}{x}y = 4x^3y^4$$

into the linear differential equation

(a) $\frac{dw}{dx} - \frac{3}{x}w = 6x^3$

(b) $\frac{dw}{dx} - \frac{6}{x}w = -12x^3$

(c) $\frac{dw}{dx} - \frac{2}{x}w = 4x^3$

(d) $\frac{dw}{dx} - \frac{6}{x}w = 6x^3$

(e) $\frac{dw}{dx} - \frac{3}{x}w = -12x^3$

9. The general solution of the differential equation

$$x\frac{dy}{dx} = y + 4x\,e^{-2y/x}$$

for $x \neq 0$ is given by

(a) $e^{2x/y} + 4\ln|x| = c$

(b) $e^{2y/x} - 4\ln|x| = c$

(c) $e^{2x/y} + 8\ln|x| = c$

(d) $e^{2y/x} - 8\ln|x| = c$

(e) $e^{2y/x} - \ln|x| = c$

10. The solution of the initial-value problem

$$(2-\cos x)\,y' + y\sin x = 2+\cos x, \qquad y(0)=0$$

is

(a) $y = \dfrac{x+\sin x}{2-\cos x}$

(b) $y = \dfrac{x+\cos x}{2+\cos x}$

(c) $y = \dfrac{2x+\cos x}{2+\cos x}$

(d) $y = \dfrac{2x+\sin x}{2-\cos x}$

(e) $y = \dfrac{2x+\sin x}{2x-\cos x}$

11. The sum of all real values of m that make the equation

$$\left(mx^2y + 2xy - y^3\right)dx + \left(x^2 - 3xy^2 + \frac{1}{3}m^3x^3\right)dy = 0$$

an exact differential equation is equal to

(a) -2

(b) 0

(c) 1

(d) -1

(e) 2

12. Which one of the following statements is true about the subset V of $\mathbb{R}^3$ defined by

$$V = \{(x_1, x_2, x_3) : x_3 = x_1 + x_2\}\,?$$

(a) V is not closed under addition but closed under multiplication by scalar

(b) V is closed under addition but not closed under multiplication by scalars

(c) V is not closed under addition

(d) V is not closed under addition but not closed under multiplication by scalars

(e) V is a subspace of $\mathbb{R}^3$

13. The general solution of the differential equation

$$\frac{dy}{dx} = \frac{x+y+2}{2x+2y-1}$$

is given by

(a) $2x + 2y - \frac{5}{3}\ln|2x + 2y - 1| = c$

(b) $x + y + \frac{5}{3}\ln|3x + 3y - 1| = c$

(c) $2y - x - \frac{5}{3}\ln|3x + 3y + 1| = c$

(d) $2x + y + \frac{5}{3}\ln|3x + 3y - 1| = c$

(e) $2y + x - \frac{5}{3}\ln|3x + 3y + 1| = c$

14. A particle is moving in a straight line with acceleration $a(t) = \ln(5t)$ and an initial velocity $v\left(\frac{1}{5}\right) = \frac{4}{5}$. The velocity at any time $t > \frac{1}{5}$ is given by

(a) $v(t) = 3t\ln(5t) - t + 1$

(b) $v(t) = 2t\ln(5t) - t + 1$

(c) $v(t) = t\ln(5t) + 2t + \frac{2}{5}$

(d) $v(t) = t\ln(5t) - t + 1$

(e) $v(t) = t\ln(5t) - 2t + \frac{6}{5}$

15. By using a suitable substitution, we can transform the differential equation

$$yy'' + (y')^2 = 0$$

into the separable differential equation

(a) $\frac{dp}{p} - \frac{dy}{3y} = 0$

(b) $\frac{dp}{p} + \frac{dy}{y} = 0$

(c) $\frac{dp}{2p} + \frac{dy}{y} = 0$

(d) $\frac{dp}{2p} - \frac{dy}{y} = 0$

(e) $\frac{dp}{p} + \frac{dy}{2y} = 0$

Answer Key

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Q	Ans	Brief explanation
1	**(e)**	Row reduce: solutions $t(-6,4,-3,1)$, so $\frac{a}{c}=\frac{-6}{-3}=2$.
2	**(a)**	$\det = 2k^2-12k-20=0 \Rightarrow k=3\pm\sqrt{19}$, sum $=6$.
3	**(a)**	$k=\frac{\ln 2}{20}$; 20% remains: $e^{-kt}=\frac{1}{5}$, so $t=\frac{20\ln 5}{\ln 2}$.
4	**(b)**	Exact: $F=\int M\,dx = x^2+ye^{xy}+g(y)$, $g'=0$, so $ye^{xy}+x^2=c$.
5	**(a)**	$m^2-6m-7=(m-7)(m+1)=0$, so $m=7$ and $m=-1$.
6	**(a)**	Highest derivative present is y''', so the order is 3.
7	**(d)**	Separable: $\sin^{-1}y=-\sqrt{1-x^2}+c$, so $y=\sin(c-\sqrt{1-x^2})$.
8	**(b)**	Bernoulli $n=4$, $w=y^{-3}$: multiply by $-3y^{-4}$ to get $w'-\frac{6}{x}w=-12x^3$.
9	**(d)**	Homogeneous, $v=y/x$: $e^{2v}dv=\frac{4}{x}dx$, $\frac{1}{2}e^{2v}=4\ln\lvert x\rvert+C$: $e^{2y/x}-8\ln\lvert x\rvert=c$.
10	**(d)**	$\rho=2-\cos x$: $((2-\cos x)y)'=2+\cos x$, $C=0$, so $y=\frac{2x+\sin x}{2-\cos x}$.
11	**(b)**	$M_y=N_x \Rightarrow m=m^3$, so $m=0,\pm1$ and the sum is 0.
12	**(e)**	$x_3-x_1-x_2=0$ is homogeneous linear: V is closed under both operations, a subspace.
13	**(c)**	$v=x+y$: $v'=\frac{3v+1}{2v-1}$, $\frac{2}{3}v-\frac{5}{9}\ln\lvert 3v+1\rvert=x+C$, times 3.
14	**(d)**	$v=\int\ln(5t)\,dt=t\ln(5t)-t+C$, and $v(\frac{1}{5})=\frac{4}{5}$ gives $C=1$.
15	**(b)**	$p=y'$, $y''=p\frac{dp}{dy}$: $yp\frac{dp}{dy}+p^2=0$, so $\frac{dp}{p}+\frac{dy}{y}=0$.